

# Function-Based and Reverse RSA Encryption Systems

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## Abstract

I have written a paper about Riemann's Hypothesis and I very consciously know that if proven, RH would have some strong implications on cryptography and cybersecurity, possibly catastrophic since RH is deeply connected to the distribution of primes — the single most crucial aspect of RSA.

So that's why in this modest paper I propose **two new encryption systems** that may or may not replace the established RSA: the *reverse-RSA* (*rRSA*) and *function-RSA* (*fRSA*).

Both come with **three variants each** — standard, super, and hybrid — making a total of **six powerful encryption methods** that improve security, flexibility, and resistance to future quantum attacks.

## Introduction

RSA's classical security is based on the difficulty of factoring the product of two large primes. However, with potential advances in quantum computing and the mathematical insights RH might bring, this foundation faces possible collapse. The purpose of this paper is to rethink RSA from the ground up, keeping the strengths of prime numbers but radically changing the role of the private and public keys by introducing **function-based transformations** and **key-hiding mechanisms**.

The *fRSA* keeps the large primes secret but transforms them via a private function before producing the public key. The *rRSA* flips this idea: it makes the primes public but hides the private function that transforms them. Both aim to expand the “hardness” beyond just prime factorization — adding layers of mathematical and computational complexity that future attacks, even quantum, would struggle to crack.

## Part 1: Function-Based RSA (fRSA)

### 1.1 Overview

fRSA is built on the idea that instead of publishing  $N = p \times q$  directly, you publish the product of their images under a secret function  $f$ . This function  $f$  is private, carefully constructed with real coefficients, and can be changed periodically to keep hackers guessing.

## 1.2 Key Generation and Public Key

- Select two large primes  $a, b$  (both private)
- Generate a unique, random, hard-to-guess function  $f$  with real coefficients
- Compute public key:

$$K_{public} = f(a) \times f(b)$$

$f(a)$  and  $f(b)$  are the images of the primes under  $f$ .

## 1.3 Security Benefits

- The function  $f$  introduces **infinite possibilities** — even a simple linear function with a random real coefficient becomes almost impossible to guess.
- If an attacker discovers one or both primes, they still need the exact function  $f$  to decrypt messages.
- Makes quantum attacks based on factoring irrelevant since the public key is no longer a direct product of primes.

## 1.4 Variants of fRSA

### 1.4.1 Standard fRSA

The function  $f$  is generated and **securely exchanged** once using traditional RSA or another secure channel. Remains fixed until manually changed.

### 1.4.2 Super fRSA

The function  $f$  is **deterministically generated at both ends** based on a shared seed or session parameters — no direct transmission of  $f$  itself.  $f$  can be changed per session or monthly, increasing security exponentially. Requires careful synchronization logic but removes the risk of  $f$  interception.

### 1.4.3 Hybrid fRSA

Combines the above: a seed or “function-generation rule” is exchanged securely once, then used to generate  $f$  deterministically on both sides. Offers flexibility and minimal transmission with maximum security.

## Part 2: Reverse RSA (rRSA)

### 2.1 Overview

rRSA flips fRSA on its head: instead of hiding the primes, it **publishes  $a, b$  openly**, but keeps the function  $f$  private. The secret key is no longer the primes but the **transformation function  $f$**  itself.

## 2.2 Key Generation and Private Key

- Public keys:  $a, b$  (primes)
- Private key: the function  $f$
- Compute “secret key” as:

$$K_{secret} = f(a) \times f(b)$$

Messages can be decrypted only if  $f$  is known.

## 2.3 Security Benefits

- Knowing  $a, b$  does not help attackers without  $f$ .
- $f$  can be complex, composed of polynomial, logarithmic, and trigonometric parts with random coefficients.
- Even partial knowledge of  $f$  is useless without the full function.

## 2.4 Variants of rRSA

### 2.4.1 Standard rRSA

$f$  is generated privately and shared securely once.  $a, b$  public.

### 2.4.2 Super rRSA

$f$  is generated deterministically from shared metadata or seed, no transmission needed. Function changes regularly, increasing security.

### 2.4.3 Hybrid rRSA

Seed or function-generation rule is exchanged securely once, then used to generate  $f$  locally each session.

## Part 3: Practical Considerations and Future Work

- Both systems require synchronization of  $f$  between sender and receiver.
- Implementation of encoding/decoding for real-valued keys and messages needs development.
- These systems open new research pathways into **post-quantum cryptography** beyond integer factorization.
- Potential for application in blockchain, secure voting, and high-surveillance environments.

## Conclusion

fRSA and rRSA, with their multiple variants, represent a novel approach to cryptography that moves beyond the limitations of classical RSA. By hiding either the primes or the function, and by leveraging function-based transformations, these systems add layers of complexity designed to thwart even future quantum adversaries. With these six protocols, we have a robust arsenal ready for the post-Riemann, post-quantum era.

**This is the future.**